USCrash: A Harmonized U.S. Nationwide Traffic Crash Dataset

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Abstract

Crash data are fundamental to traffic safety research for understanding crash mechanisms and developing effective safety models. However, existing crash datasets are often limited in geographic coverage, crash types, data scale, and attribute consistency. To address these limitations, we developed **USCrash**, a large-scale harmonized traffic crash dataset constructed from official state crash records across the United States. USCrash contains over 65.5 million crash-level records from 41 states and the District of Columbia spanning 2001–2026. Heterogeneous state-specific variables are harmonized into a common framework of 77 MMUCC-aligned attributes covering temporal, spatial, severity, collision, environmental, roadway, behavioral, and provenance information. Validation against official crash counts across main U.S. cities and counties shows close agreement, with an aggregate difference of only 0.08%. With its broad geographic and temporal coverage, rich crash-level attributes, and high consistency with official records, USCrash provides a unified and reproducible data foundation for potential nationwide traffic safety analysis, multi-scale crash modeling, and emerging AI-based safety applications.

Keywords

Traffic safety, Crash data, Data harmonization

1 Introduction

Crash data are fundamental to traffic safety research because they provide information on when and where crashes occur, the types of crashes involved, their severity, and other critical characteristics. These data enable researchers to identify spatial and temporal crash patterns, investigate underlying crash mechanisms, and thereby develop countermeasures to improve traffic safety. Recognizing the critical value of crash data for public safety analysis, many countries and agencies have established systems for collecting and disseminating crash records (e.g., UK's STATS19 dataset [1]; France's BAAC dataset [2] and New Zealand's CAS dataset[3]).

However, the United States still lacks a public, harmonized nationwide crash database. Although NHTSA has promoted greater consistency in crash reporting through initiatives such as the Model Minimum Uniform Crash Criteria (MMUCC) [4], crash records are ultimately collected, maintained, and disseminated by individual states [5, 6]. As a result, states vary widely in crash data availability, variable definitions, coding practices, and access formats, ranging from downloadable datasets to dashboard systems as reported in [7]. These inconsistencies make it difficult and time-consuming for researchers and the public to acquire data and conduct reproducible cross-state crash analyses using a uniform data framework.

Although several U.S. official crash datasets (e.g., FARS [8], CRSS [9], HSIS [10], and CISS [11]) and public datasets (e.g., US-Accidents [12] and ML4RoadSafety [13]) have provided open-source large-scale crash data, they are either old, limited to specific crash types, geographic scope, and crash attributes, as shown in Table 1. To address these limitations, we present **USCrash**, a harmonized U.S. nationwide crash dataset comprising over 65 million records, with harmonized state-specific records into 77 MMUCC-guided attributes. Its main contributions are:

- **Broad coverage and large scale:** USCrash integrates official crash records from 41 states and DC, including all types of crashes for a total of 65.54 million records.
- **Cross-state semantic harmonization:** Heterogeneous state-specific variable systems are harmonized into 8 major crash attribute groups with 77 MMUCC-guided features, while preserving clear source identifiers and mapping rules.
- **High reliability and reproducibility:** Official data sources, mapping provenance, and cross-city validation ensure its reliability and reproducibility than existing open-source datasets.

The rest of the paper is organized as follows. Section 2 describes the dataset creation processing, and Section 3 demonstrates the dataset statistics, validation procedures, and key insights. Section 4 discusses its potential applications and the dataset limitations. Section 5 concludes the paper.

Table 1: Comparison of major U.S. crash data resources.

Dataset	Source	Geography & years	Crash types	Scale	Data integration & features
FARS	National fatal-crash census	Nationwide; 1975–2024	Fatal crashes only	1.87M	Standard federal schema with linked crash, vehicle, and person data; public
CRSS	National sampling	Nationwide; 2016–2024	Fatal, injury, and PDO crashes	0.43M	Standardized survey data and weights for national estimates
HSIS	Official state records	7 states; 1985–2024	Fatal, injury, and PDO crashes	~5.0M	Harmonized crash, roadway, and traffic data (request-based access)
CISS	In-depth crash sample	Nationwide; 2017–2024	Tow-away vehicle crashes	0.027M	Detailed crash, vehicle, and injury investigations
US-Accidents	Third-party feeds	49 states; 2016–2023	Unknown	7.73M	Common event representation; severity reflects traffic impact rather than injury
USCrash	**Official state records**	**41 states + DC; 2001–2026**	**Fatal, injury, and PDO crashes**	**65.54M**	**77 MMUCC-aligned harmonized attributes covering temporal, spatial, severity, environmental, and other details.**

2 Dataset Creation Process

USCrash is constructed in three stages: data source acquisition, crash-level data preparation, and MMUCC-guided semantic harmonization (Fig. 1). The workflow transforms heterogeneous state crash records into a common representation while retaining source-specific crash reporting scope and detail.

2.1 Data sources and acquisition

Crash data were assembled from multiple state transportation agencies, public data portals, and other archived data-release channels. As these data are disseminated differently across states and years, we collected crash records through four main acquisition approaches according to the data availability:

(1) Direct data downloading. Several state DOTs have published their crash data as downloadable files (e.g., CSV, Excel, ZIP, and Geodatabase files), so we retrieved the available annual or multi-year releases directly from their websites and associated repositories. Examples include annual crash files published by the Colorado DOT and Pennsylvania DOT, California CHP/CCRS annual files, and Wisconsin's downloadable DT4000 archives.

(2) Public data services/APIs. Many states expose crash records through public data services rather than file downloads. These include ArcGIS-based endpoints, Socrata APIs, and other structured open-data interfaces. We programmatically queried these services from states such as Alaska, Arkansas, Connecticut, Delaware, Iowa, Illinois, Massachusetts, New York, and others.

(3) Web-based query and export systems. Some states provide crash records through public websites, dashboards, or query tools that allow users to search and export records but do not provide public data files or API. For these systems, records were obtained through the site's available query and export functions, either manually or through scripts that reproduced the public workflow. Examples include Maine's Public Crash Query Tool, Georgia's crash dashboard, and Nebraska's NTIP system.

(4) Formal agency requests. For crash records not available through public data download/services or web-based export systems, we obtained files through official agency request procedures. Requested datasets include records from Arizona, Texas, Wyoming, and portions of Nevada and Washington. These files were processed according to the access, use, and redistribution conditions specified by the providing agencies (e.g., omitted sensitive features).

Acquisition methods sometimes varied across years within the same state as data availability and dissemination changed over time. We therefore combined multiple official sources where necessary to maximize temporal coverage and obtain the most complete crash-level records available. This process yielded harmonized crash data for 41 states and the District of Columbia. Detailed source inventories, acquisition methods, temporal coverage, and access conditions are provided in Supplementary Note 1.

2.2 Crash-level data preparation

After acquisition, all source data were converted into a consistent crash-level structure with each row representing a unique crash record for subsequent harmonization. Some state files already contain one record per crash, while others store crash, vehicle, person, or event information in separate tables. These tables were linked using source crash identifiers/keys, and supported information was aggregated to the crash level. For example, if a crash involves several vehicles, pedestrians and cyclists, corresponding attributes such as vehicle count, fatalities and injuries, and road user involvement were summarized from the linked records.

Meanwhile, the same crash may appear in multiple download batches or overlapping source files. We therefore checked crash identifiers within the appropriate state and year context and reconciled overlapping records using source-specific rules to avoid duplicate counting, combining complementary information where supported. State-specific inclusion criteria were applied only when documented by the source, and detailed preparation procedures and quality-control checks are summarized in Supplementary Note 2.

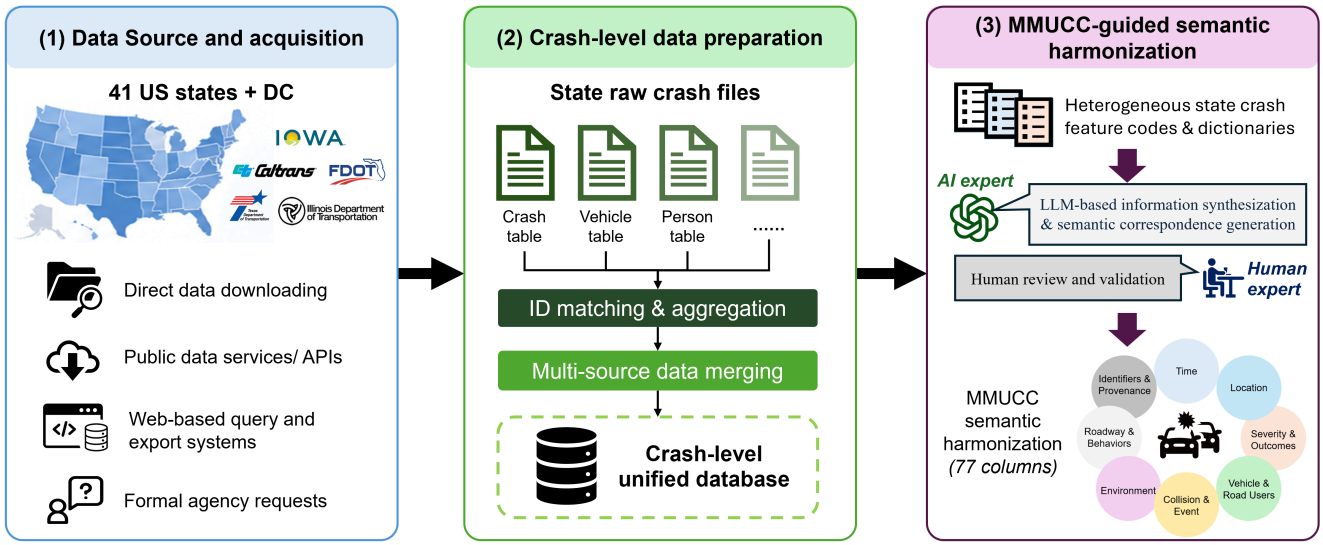


Figure 1: USCrash dataset construction workflow.

2.3 MMUCC-guided semantic harmonization

Crash records differ substantially across states in both the attributes and the way similar attributes are defined and coded. For example, some states record information that is absent in others, while conceptually similar variables may use different field names, category systems, or definitions. These inconsistencies make it hard to directly combine state datasets into a common analytic framework. We therefore harmonized the heterogeneous state records into a common representation using the MMUCC by NHTSA as the primary semantic reference [4].

Given the large volume and heterogeneity of source schema and documentation across states and years, we leveraged LLM to efficiently synthesize this information and identify candidate semantic correspondences, followed by human expert review and validation to establish the final semantic mappings. Specifically, GPT-6 Astra [14] was used to review state data dictionaries, crash-report manuals, codebooks, and field descriptions; organize source attributes; compare variable definitions and coded values; and generate candidate cross-state mappings aligned with MMUCC concepts. The candidate mappings were then manually reviewed and validated by the human experts against the original state documentation, with ambiguous or potentially non-equivalent concepts examined individually before inclusion in the final mapping framework.

Finally, the mappings generated a harmonized 77-column schema (Fig. 2), organizing crash information into eight major groups: **identifiers and provenance, time, location, severity and outcomes, vehicles and road users, collision and event characteristics, environment conditions, and roadway and behavior context**. Specifically, the dataset includes basic crash information such as crash date and time, geographic coordinates and jurisdiction, KABCO severity level, fatality and injury counts, vehicle counts, and indicators of involved road users. It also captures detailed crash context such as collision type/manner, first harmful event, weather, lighting, road-surface condition, alcohol and drug involvement, speeding, hit-and-run, posted speed limit, and other information where available. Raw source codes, labels, source identifiers, and provenance fields are also retained alongside the harmonized attributes to preserve traceability. When a source does not provide sufficient information for a reliable mapping, the corresponding value is retained as unknown, unavailable, or source-specific rather than inferred. Detailed field definitions and state-specific mapping rules are provided in Supplementary Notes 3.

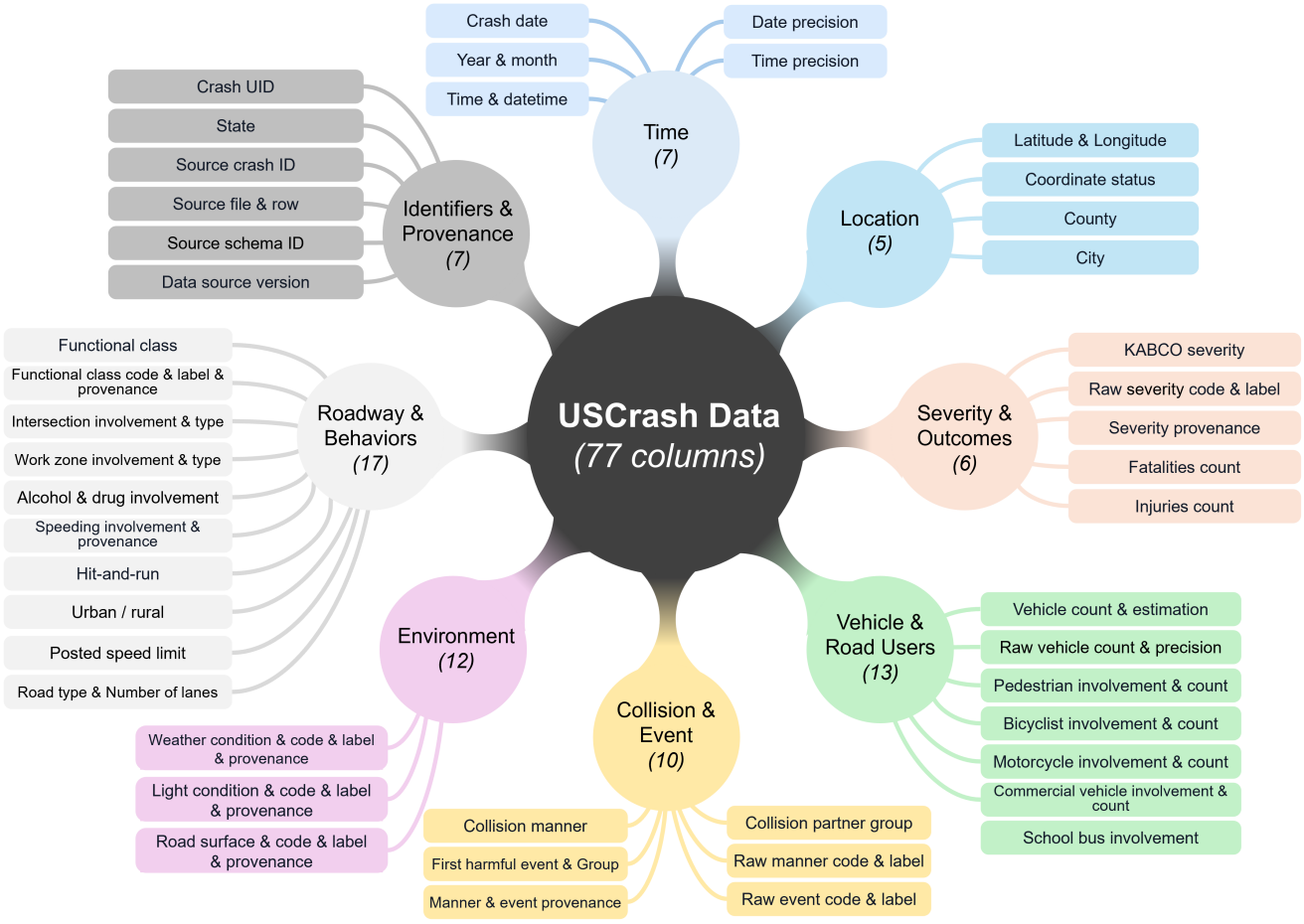


Figure 2: USCrash common schema and representative attributes.

3 USCrash Dataset

3.1 Spatial-temporal and attribute coverage

USCrash dataset comprises 65,541,414 crash-level records across most U.S. states with substantial variation in spatial, temporal, and attribute coverage across jurisdictions:

(1) Spatial distribution. Fig. 3(a) illustrates the spatial distribution of crashes in both the national (5 km × 5 km grids) and representative city levels (250 m × 250 m grids) using 2021 as a common reference year. The national distribution shows that crash records cover most major urban areas, with dense concentrations around metropolitan regions and along major transportation corridors. The four city-level examples—Seattle, Chicago, Los Angeles, and Houston—further illustrate the finer spatial detail available in urban areas, including recognizable street-grid structures and localized high-density corridors. Overall, these patterns demonstrate the broad geographic coverage of our dataset spanning both nationwide transportation networks and diverse urban environments.

(2) Temporal coverage. The USCrash dataset spans 2001–2026 across 511 state–year files, with an average of 12 available years per state (including DC; Fig. 3b). Most crash records are concentrated in 2016–2025, which accounts for 72.9% of all records. Coverage peaks in 2021, with approximately 5.94 million records across 41 states. Massachusetts and New Jersey provide the longest continuous temporal coverage, each contributing 26 annual files from 2001 through 2026, whereas New Mexico has the shortest coverage, with records available only for 2019. In particular, the dataset spans a substantial pre-pandemic baseline, the COVID-19 pandemic period, and subsequent years, enabling longitudinal comparisons of crash patterns across these periods. For example, 34 states contain records in both 2019–2023, supporting within-jurisdiction comparisons before and during the pandemic.

(3) Attribute availability. Fig. 3(c) summarizes attribute availability across the eight common-schema families, revealing substantial cross-state heterogeneity in the information recorded in state crash datasets. At the attribute level, identifier and provenance fields and crash time are nearly complete, while availability is highest for severity and outcomes (87.1%), environmental conditions (86.1%), and location (81.6%). Collision and event characteristics reach 66.7%, whereas vehicle and road-user attributes and roadway and behavioral context reach 48.3% and 30.7%, respectively. Across states, Texas has the highest overall attribute availability (93.3%), followed by South Dakota (90.0%) and Virginia (88.6%), while Maine (31.2%), Nebraska (34.5%), and Idaho (38.6%) show the lowest levels. Overall, these substantial differences reflect variation in state-level crash-data collection, coding, and reporting practices, with some states open-source substantially fewer contextual attributes than others. The observed disparities further highlight the need for a more nationally consistent and comprehensive framework for crash-data collection and reporting.

3.2 Crash data statistics

To illustrate the information captured in the collected USCrash dataset, Fig. 4 summarizes key statistics on temporal patterns, severity distribution, collision type, environment features, roadway and behavioral characteristics across the dataset.

(1) Temporal patterns. Fig. 4(a)–(c) summarizes the temporal distribution of crashes by month, day of week, and hour of day. Monthly crash shares remain relatively stable throughout the year, ranging from 7.6% to 9.3%, with the highest proportions observed from October to December. Across the week, Friday accounts for the largest share of crashes (16.7%), while Sunday has the lowest share (10.9%). The hourly distribution shows distinct weekday and weekend patterns. Weekday crashes exhibit pronounced peaks during the morning commuting period (7:00–9:00) and afternoon peak period (16:00–17:00), whereas weekend crashes are more broadly distributed from midday through the evening and show relatively higher proportions during late-night and early-morning hours.

(2) Severity distribution. Fig. 4(d) presents the harmonized KABCO severity distribution. No-injury crashes (O) account for the largest share at 69.6%, followed by possible-injury crashes (C) at 11.2% and minor-injury crashes (B) at 8.5%. Serious-injury crashes (A) account for 2.2%, while fatal crashes (K) represent 0.5%. Unknown/unresolved severity (U) accounts for 8.1% of records with missing severity values in the crash records.

(3) Crash type and manner. Fig. 4(e) shows the distribution of crash types and collision manners. Rear-end crashes form the largest crash type category, accounting for 24.6% of all records, followed by crashes not involving a collision with a vehicle in transport (22.2%) and angle crashes (19.3%). Sideswipe crashes in the same direction account for 8.9%, while head-on crashes and opposite-direction sideswipes account for 1.7% and 1.2%, respectively. Due to data limitation, 14.4% of the crashes are classified as unknown or unavailable. Within crashes not involving a vehicle in transport, fixed-object collisions form the largest group (22.2%), followed by parked-vehicle collisions (9.3%), and animal collisions (6.6%). In contrast, pedestrian crashes (3.1%), rollover or overturn events (2.1%), and bicyclist or other non-motorist (1.0%) crashes account for relatively small portions.

(4) Environment features. Fig. 4(f) summarizes the joint distribution of weather, lighting, and road-surface conditions. Most crashes occur under clear weather (75.2%), daylight (67.8%), and dry road-surface conditions (80.5%). Cloudy (10.7%) and rainy (9.3%) conditions are also relatively common, while dark-lighted (17.8%) and dark-unlighted (9.8%) conditions represent the main non-daylight categories. Wet roads account for 14.6% of crashes, whereas snow, ice/frost, and other surface conditions each represent relatively small proportions. The linked flows further show the joint combinations of these environment features, reflecting the diverse environment scenarios of the crash records.

(5) Roadway and behavioral features. Fig. 4(g)–(i) summarizes roadway functional class, posted speed limit, and selected behavioral indicators. Principal arterials account for the largest share of crashes (26.5%), followed by interstates (21.2%), minor arterials (19.4%), and major collectors (12.5%). Posted speed limits are concentrated in the 26–35 mph (28.8%) and 36–45 mph (29.2%) ranges, followed by 46–55 mph (15.2%), 56–65 mph (9.8%), ≤ 25 mph (9.2%), and 66–85 mph (7.7%). Among the selected behavioral indicators, speeding is identified in 5.0% of crashes, alcohol involvement in 2.4%, drug involvement in 0.5%, and hit-and-run in 9.3%. However, a substantial proportion of records do not contain sufficient information to determine these behavioral factors.

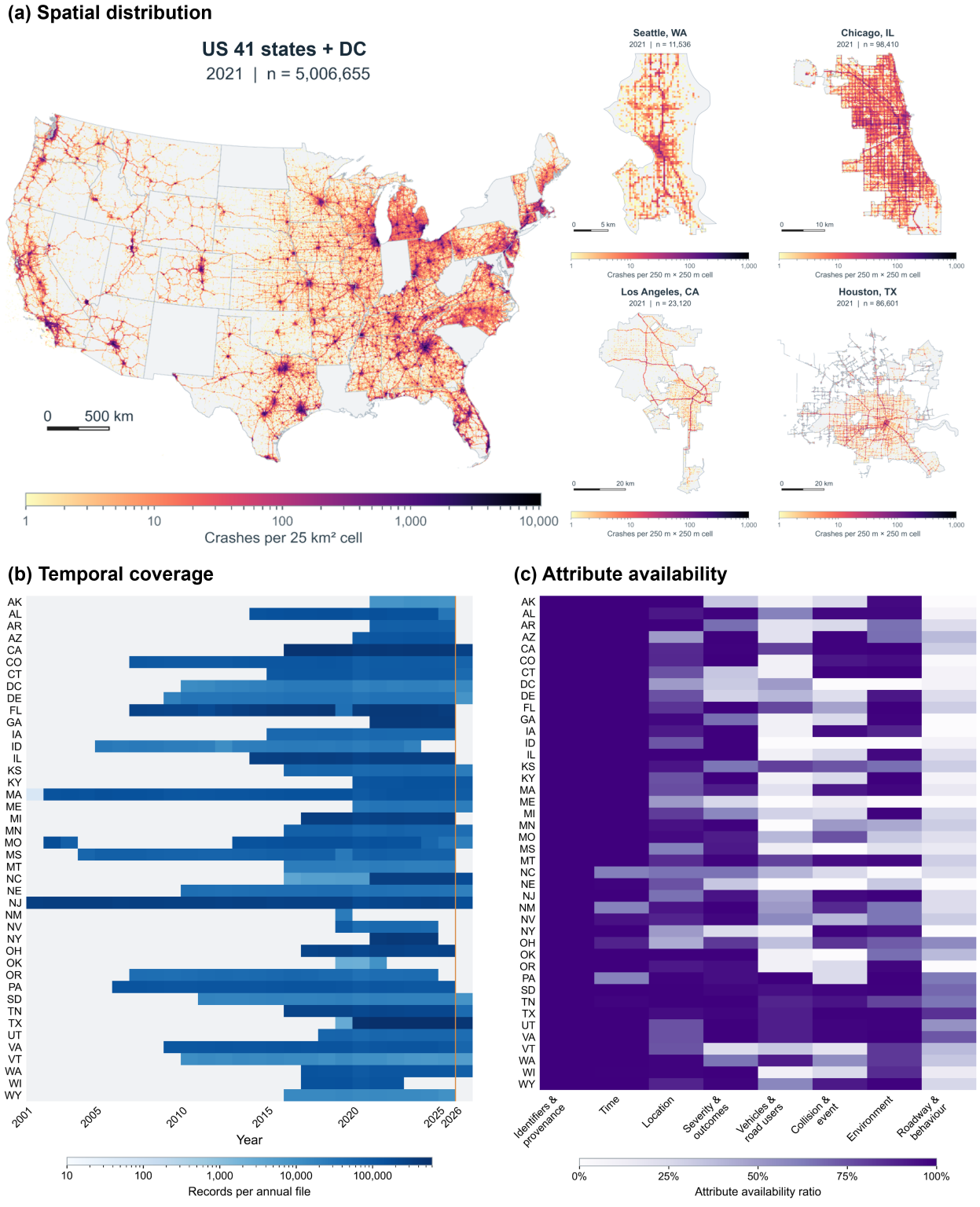


Figure 3: Spatial, temporal, and attribute coverage of USCrash dataset.

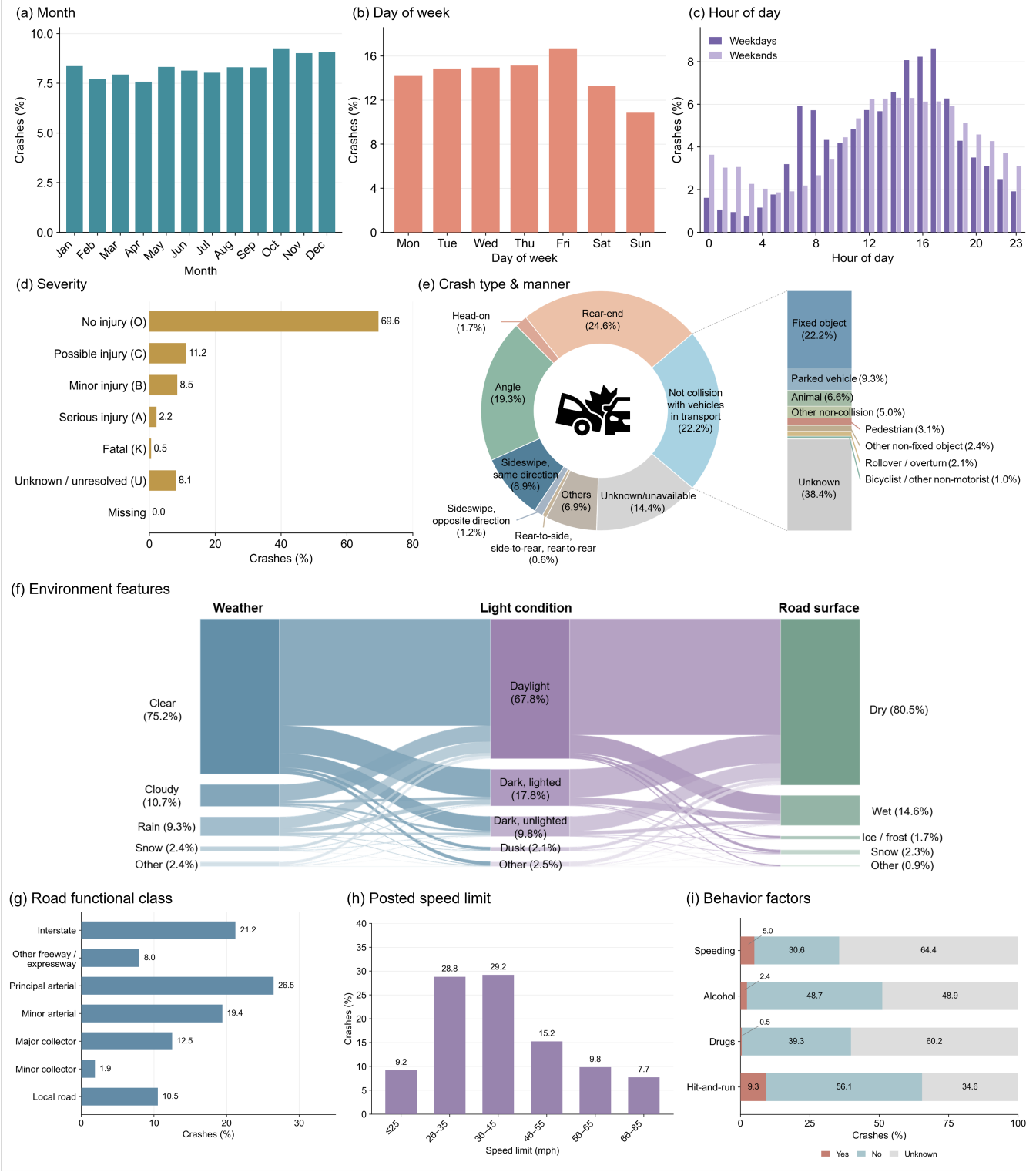


Figure 4: Crash statistics derived from the harmonized USCrash attributes.

Table 2: Comparison of USCrash and US-Accidents with official crash counts for 20 U.S. cities and counties

Area	Official	USCrash	Diff.	US-Accidents	Diff.	Area	Official	USCrash	Diff.	US-Accidents	Diff.
Phoenix, AZ	40,833	40,876	+0.11%	9,348	-77.11%	Detroit, MI	25,147	25,147	0.00%	3,438	-86.33%
Los Angeles, CA	29,953	30,033	+0.27%	33,742	+12.65%	Kansas City, MO	18,283	18,280	-0.02%	4,324	-76.35%
San Diego, CA	8,338	8,332	-0.07%	11,014	+32.09%	Portland, OR	6,430	6,430	0.00%	5,128	-20.25%
San Francisco, CA	5,805	5,767	-0.65%	2,929	-49.54%	Philadelphia, PA	10,417	10,552	+1.30%	4,383	-57.92%
Hillsborough, FL	28,020	28,019	0.00%	11,581	-58.67%	Austin, TX	12,465	12,571	+0.85%	12,417	-0.39%
Orange, FL	27,422	27,434	+0.04%	36,576	+33.38%	Dallas, TX	35,366	35,326	-0.11%	26,037	-26.38%
Atlanta, GA	37,208	37,080	-0.34%	4,306	-88.43%	Houston, TX	67,045	66,887	-0.24%	28,245	-57.87%
Aurora, IL	3,754	3,757	+0.08%	20	-99.47%	San Antonio, TX	38,467	38,852	+1.00%	5,747	-85.06%
Chicago, IL	98,563	98,570	+0.01%	5,601	-94.32%	Richmond, VA	5,656	5,666	+0.18%	4,008	-29.14%
Boston, MA	4,139	4,139	0.00%	3,207	-22.52%	Seattle, WA	12,106	12,106	0.00%	3,542	-70.74%
Total	**284,035**	**284,007**	**-0.01%**	**118,324**	**-58.34%**	**Total**	**231,382**	**231,817**	**+0.19%**	**97,269**	**-57.96%**

3.3 Dataset verification

To verify the consistency and reliability of USCrash dataset, we compared the crash counts with corresponding official records[1] for 20 representative U.S. cities and counties as shown in Table 2. The selected areas span multiple states and a broad range of urban scales. For comparison, crash counts from the US-Accidents dataset were evaluated for the same geographic areas and reference year. The results demonstrate clearly that:

USCrash closely aligns with official crashes across all cities. The percentage differences range from −0.65% in San Francisco to +1.30% in Philadelphia, with several areas showing exact or near-exact agreement (e.g., Boston, Chicago, Detroit, Portland, and Seattle). Across all selected areas, the official records contain 515,417 crashes, compared with 515,824 records in USCrash, corresponding to an overall difference of only +0.08%. This close agreement is consistently observed across different states and cities, indicating that the crash populations retained in USCrash closely match the corresponding official records.

In contrast, US-Accidents shows substantial discrepancies from official crash counts. It contains 215,593 records across the same areas, approximately 58.17% fewer than the official total. The magnitude of the discrepancy varies considerably by location, ranging from large undercounts in Aurora (−99.47%), Chicago (−94.32%), and Atlanta (−88.43%) to counts exceeding the official totals in Orange County (+33.38%), San Diego (+32.09%), and Los Angeles (+12.65%). These discrepancies largely stem from the use of third-party traffic-event feeds, such as MapQuest and Microsoft Bing, which are based on user- and service-generated reports rather than official crash records and can therefore lead to substantial under- or overreporting [12].

Fig. 5 further show the difference between the two datasets, providing strong empirical support for the reliability of USCrash dataset. Meanwhile, USCrash retains rich crash-level information on temporal, spatial, severity, collision, environmental, roadway, and behavioral characteristics, which are largely absent from existing large-scale public datasets such as US-Accidents.

[1]Official crash counts were obtained from corresponding government websites, reports, and other official documents. An expanded comparison across more U.S. cities is provided in Supplementary Note 4.

4 Dataset applications and limitations

The broad spatial coverage, long temporal span, and rich harmonized crash attributes of USCrash dataset enable multiple potential traffic safety applications:

- **Macroscopic traffic safety analysis:** USCrash can be integrated with macroscopic traffic exposure measures, such as annual average daily traffic (AADT) and socioeconomic indicators, to support crash-frequency modeling and safety performance function development at national, state, and regional scales [15].
- **Mesoscopic network-level crash modeling:** USCrash can be linked with roadway networks, including arterials and freeways, as well as intersections, corridors, and associated traffic characteristics, to support network-level spatiotemporal crash modeling and safety assessment [16, 17].
- **Microscopic crash risk and severity prediction:** USCrash can be combined with high-resolution traffic flow and operation data and weather data for real-time crash-risk prediction [18], while individual crash records can support crash-causation analysis and injury-severity modeling [19].
- **AI-based safety applications:** The harmonized data structure provides a foundation for emerging AI-based safety applications, including the integration of large language models for crash-scene interpretation and contextual reasoning [20], as well as generative models for critical-scenario generation to support both human-driven and automated vehicle safety [21].

However, two main limitations remain. First, USCrash primarily consists of structured and coded crash attributes and generally does not contain detailed narrative descriptions and other contextual information (e.g., image and video) that could support more comprehensive reconstruction and interpretation of individual crashes. Second, the accuracy of reported crash time and location depends on the original state records and reporting processes. In some cases, temporal or spatial information may contain reporting delays, geocoding errors, or other discrepancies inherited from the source data [22, 23]. More detailed source information is not consistently available across jurisdictions to correct the current dataset. Future extensions that incorporate richer crash narratives, multimedia information, and other contextual data could further expand the analytical capabilities of USCrash.

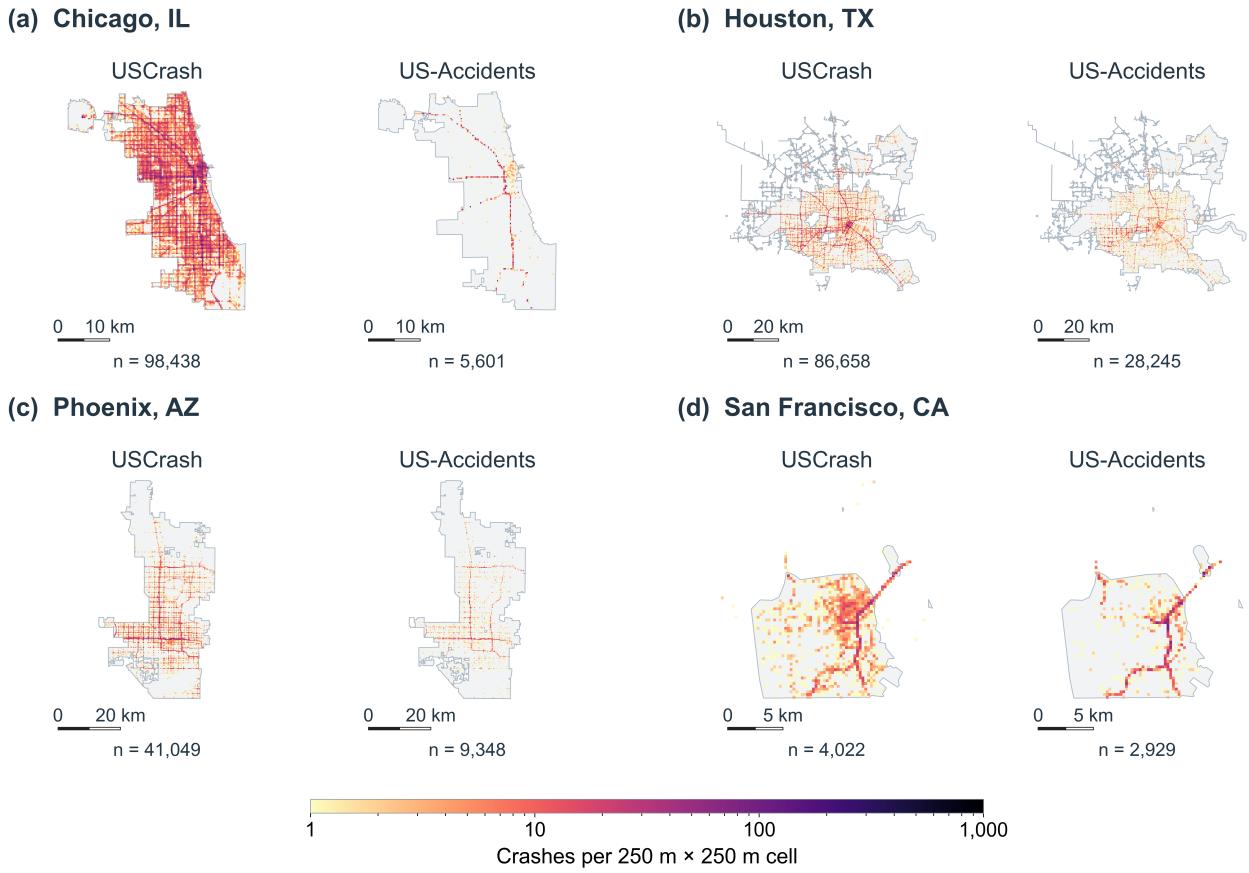


Figure 5: Comparison of USCrash and US-Accidents in 4 U.S. cities.

5 Conclusion

This paper presents USCrash, a harmonized nationwide traffic crash dataset containing over 65 million crash-level records from 41 states and the District of Columbia across 2001–2026. Heterogeneous state records are standardized into 77 MMUCC-aligned attributes covering temporal, spatial, severity, collision, environmental, roadway, and behavioral information while retaining source and mapping provenance. Validation with official crash records across U.S. cities and counties shows close agreement, supporting the reliability of the dataset. With its broad spatial-temporal coverage, rich crash-level details, and reproducible harmonization framework, USCrash provides a unified data foundation for traffic safety analysis, crash modeling, and emerging AI-based safety applications.

6 Data and Code Availability

USCrash is a released in XX; Data organization and representative-year reproducibility results are described in Supplementary Note 5.

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